Montaser Mohammedalamen, PhD

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Work Experience

- **Alberta Machine Intelligence Institute (Amii)** — *July 2025 - Present*
  Applied Research Scientist, AI Trust & Safety — Edmonton, Canada
  - **Research Strategy:** Establishing Amii's safety research dual-track strategy using RL to risk assignment generative models, and mitigating structural RL safety (uncertainty, non-stationarity, partial reward observability).
  - **Technical Collaboration:** Partnering with Cohere, and national institutes Canadian AI Safety Institute (CAISI), CIFAR Medical AI Working Group, Mila, the Vector Institute, and the National Research Council (NRC).
  - **Academic Outreach:** Directed a 10-session technical AI safety track for Amii's annual Upper Bound conference, serving an audience of over 11,000 attendees.
  - **Team Building:** Hired a Research Scientist and an ML Engineer for the team while overhauling the technical interviews framework now adopted by Amii.

- **SonyAI** — *June 2020 - Oct 2020, Jan 2021 - Oct 2022*
  AI Engineer — Tokyo, Japan
  - **Robotics Research:** Co-developed a table-tennis robot on the ACE Team capable of defeating professional players, resulting in a Nature publication.
  - **RL Policy:** Trained RL policies for targeted, high-precision ball returns and led the technical sub-team for robot serving.
  - **Sim-to-Real & System Integration:** Built multi-agent simulation environments using self-play and goal-conditioned RL, successfully transferring trained RL policies onto real-time robot control loops.

- **University of Alberta** — *Feb 2020 - Dec 2020*
  Research Associate — Edmonton, Canada
  - Formulated a novel robust optimization framework that enables autonomous control agents to learn cautious behaviors when encountering novel, out-of-distribution states, supervised by Prof. Michael Bowling.

- **SonyAI** — *July 2019 - Jan 2020*
  Machine Learning Intern — Tokyo, Japan
  - Design an RL environment and trained policies using imitation learning to acquire complex cooking skills from chef demonstrations, successfully deploying the models on physical robots.

Education

- **University of Alberta** — PhD in Computing Science — *Sep 2022 – Dec 2025*
  - **Thesis:** "Reinforcement Learning with Partially Observable Rewards".
  - **Advisor:** Prof. Michael Bowling

- **African Institute for Mathematical Sciences** — M.Sc. in Machine Intelligence — *Sep 2018 – Sep 2019*
  - **Performance:** Overall score 75%
  - **Advisor:** Prof. Benjamin Rosman

- **University of Khartoum** — B.Sc. (Honors) in Electronics & Computer Engineering — *Aug 2013 – Sep 2018*
  - **Performance:** First Class Honors (Ranked 3rd in the Electronics department, CGPA 7.5/10)

Selected Publications & Patents

Peer-Reviewed Journal Articles

[J.1] **Montaser Mohammedalamen**, Dustin Morrill, Alexander Sieusahai, Yash Satsangi, and Michael Bowling, "Learning to Be Cautious", *Transactions on Machine Learning Research (TMLR)*, 2025.

Refereed Conference Publications

[C.1] **Montaser Mohammedalamen** and Michael Bowling, "Generalization in Monitored Markov Decision Processes (Mon-MDPs)," *Reinforcement Learning Conference (RLC)*, Montreal, Canada, 2026.

[C.2] Simone Parisi, **Montaser Mohammedalamen**, Alireza Kazemipour, Matthew E. Taylor, and Michael Bowling, "Monitored Markov Decision Processes", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, Auckland, New Zealand, 2024.

Workshop Papers & Preprints

[W.1] **Montaser Mohammedalaman**, K. Roice, R. McLean, A. Lefaivre Škopac, "A Systematic Investigation of The RL-Jailbreaker in LLMs," *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Machine Learning (ICML)*, 2026.

[W.2] R. McLean, T. E. Lee, **Montaser Mohammedalaman**, K. Roice, G. Berseth, P. M. Pilarski, M. C. Machado, A. Lefaivre Škopac, B. Rosman, "AI Agent Safety is a Reinforcement Learning Problem", *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Machine Learning (ICML)*, 2026.

[S.1] **Montaser Mohammedalamen**, Michael Bowling, and Matthew E. Taylor, "Learning To Be Cautious for Industrial Chemical Control," *Under review*.

[W.3] D. Khamies, **Montaser Mohammedalamen**, and B. Rosman, "Transfer Learning for Prosthetics Using Imitation Learning," *Black in AI workshop, NeurIPS*, arXiv preprint 1901.04772, 2018.

Theses

[T.1] **Montaser Mohammedalamen**, "Reinforcement Learning with Partially Observable Rewards", PhD Thesis, University of Alberta, 2026.

[T.2] **Montaser Mohammedalamen**, "Learning Actions Representation In Reinforcement Learning for Safe Exploration," Master's Thesis, 2019.

Patents

[P.1] **Montaser Mohammedalamen**, "Wheelchair Robot controlled by Brain signal," Patent No. 4254, Patent Office, Ministry of Justice, Sudan, 2015-2016.

[P.2] **Montaser Mohammedalamen**, "Smart Farm," Patent No. 3334, Patent Office, Ministry of Justice, Sudan, 2015-2016.

Professional Service

- **Transactions on Machine Learning Research (TMLR)** — *2024 – Present*
  Journal Reviewer
- **International Conference on Machine Learning (ICML)** — *2026*
  Gold Reviewer
- **Neural Information Processing Systems (NeurIPS)** — *2025 & 2026*
  Conference Reviewer
- **Reinforcement Learning Conference (RLC)** — *2025 & 2026*
  Technical Reviewer
- **Adaptive and Learning Agents (ALA) Workshop at AAMAS** — *2025 & 2026*
  Workshop Co-organizer and Area Chair
- **Amii Fellows Trust & Safety Research Proposals** — *2026*
  Committee Member
- **Cisco-University of Alberta Consortium for Agentic Research (CUACAR)** — *2026*
  Research Proposal Reviewer

Teaching Experience

- **University of Alberta** *Jan 2023 – May 2025*
  Head Teaching Assistant — Experimental Mobile Robotics (COMP 412/503) Edmonton, Canada
  - Coordinated the TA team, managed grading frameworks, and oversaw lab infrastructure for a mixed cohort of senior undergraduate and graduate students.
  - Designed and developed 8 comprehensive lab modules for 35 students using the Duckietown to demonstrate physical robotic control.
  - Instructed students on core robotics principles, including basic kinematics, ROS commands, P/PD/PID controllers, computer vision integration, and ML-based traffic sign detection.
  - Mentored students through a capstone final project implementing an end-to-end autonomous driving pipeline featuring lane tracking, obstacle avoidance, and traffic rule navigation.
  - Delivered specialized guest lectures on RL in robotics alongside career development strategies for industry internships.

Selected Projects

- **RL for LLMs Risk Assessment** *Ongoing*
  Generative models Safety
  - Tech Lead, developing a three-stage optimization framework that reframes generative model jailbreaking as a sequential decision problem. The system structurally decomposes environmental and algorithmic components to isolate the core drivers of adversarial success, scales exploitation pipelines across multi-modal action spaces (text, vision, and audio), and embeds these insights into a unified two-player zero-sum game.
- **Learning to Be Cautious for Industrial Chemical Control** *Ongoing*
  Critical Applications Safety
  - Generalized the "learning to be cautious" framework to continuous action spaces, extending its mathematical applicability to high-dimensional, safety-critical settings, enabling robust control under environmental uncertainty. Validated the approach on complex chemical process control benchmarks, demonstrating practical scalability and bridging the gap between theoretical safe RL and real-world industrial deployments.
- **Plasticity Loss in Monitored MDPs** *Ongoing*
  Continual RL
  - Investigating the structural mechanisms of plasticity loss within Monitored Markov Decision Processes (Mon-MDPs) to preserve agent adaptability during continual learning. Analyzing how the dual-learning architecture, simultaneously training a predictive reward model alongside deep Q-networks, compounds drives long-term performance degradation. Formulating a systematic framework to evaluate the intrinsic vulnerabilities of Mon-MDPs to this phenomenon while adapting advanced regularization and reset mitigation strategies to maintain an optimal stability-plasticity balance.
- **Monitored Markov Decision Processes**
  Theoretical & Practical Evaluation frameworks []
  - Addressed settings where reward structures are unobservable to the agent, analyzing consequences across varied testing scenarios.
- **Learning to be Cautious**
  Robust optimization algorithms []
  - Constructed parameters enabling systems to safely deploy context-aware behavior patterns dynamically within unfamiliar boundaries.

Skills

- **Programming Languages:** Python, JAX, PyTorch, TensorFlow, C++
- **Machine Learning & RL:** Reinforcement Learning (RL), Safety & Trust, Multi-agent RL, Optimization, Imitation Learning, Generative models, LLMs safety, Jailbreaking LLMs, Game Theory, Continual RL.
- **Deep Learning Architectures:** CNN, RNN, LSTM
- **Robotics & Deployment:** Robotics, ROS, Physics Simulation, Linux, Git, LaTeX, AWS, GCD, Slurm

Honors and Awards

- **Andrew Stewart Memorial Graduate Prize** *May 2025*
  University of Alberta (5,000 CAD)
- **Graduate Student Engagement Scholarship** *Jan 2024*
  University of Alberta (10,000 CAD)
- **Graduate Student Engagement Scholarship** *Jan 2023*
  University of Alberta (10,000 CAD)
- **AMMI Master Scholarship** *Sep 2019*
  African Master in Machine Intelligence (15,000 USD)
- **Best Graduation Project ("RL for Prosthetics")** *Oct 2018*
  University of Khartoum, Sudan
- **Undergraduate Research Prize** *Dec 2017*
  University of Khartoum (2,000 USD)
- **Audience Prize** *Nov 2017*
  Falling Walls Lab Finals, Berlin, Germany
- **ICT Professional Foundation Scholarship** *Nov 2016*
  Ericsson Middle East University Program
- **IEEE Extreme 10.0 Programming Competition** *Oct 2016*
  IEEE Sudan Section Ranked 5th in Sudan, 988th of 2500 teams globally
- **Best Project Award** *2015 - 2016*
  Electrical & Electronics Engineering Students Exhibition Awarded consecutively for two years

Volunteering

- **Sudanese Machine Learning Community (SMLC)** *Dec 2019 - Present*
  Co-founder & Core Representative []
  - Co-founded an initiative building AI capacity locally, connecting resources to regional engineering candidates.
- **SMLC Mentorship Program** *Dec 2019 - Present*
  Academic Research Mentor
  - Mentoring undergraduate students at the University of Khartoum on custom baseline RL graduation items.
- **EEESE Hackathon** *Apr 2016 - Aug 2016*
  Founder & Event Coordinator
  - Established the first hackathon event setup in Sudan, coordinating 28 student researchers from 7 distinct universities.
- **Academic Committee for EEESE** *Aug 2015 - July 2016*
  Head of Committee University of Khartoum